



Pulse evolution and multi-pulse state of coherently coupled polarization domain walls in a fiber ring laser

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Abstract: Pulse evolution and multi-pulse state of coherently coupled polarization domain walls (PDW) is experimentally demonstrated in a novel fiber ring laser. Versatile pulse shapes benefit by wide range moving of PDW in the weakly birefringent fiber. The 8.6 m short-cavity structure is more compact and accessible based on a 976 nm pump with nearly zero negative dispersion (-0.0002 ps^2). Besides, multi-pulse patterns such as PDW splitting, harmonic mode-locking, and periodic soliton collision are also observed under larger net negative dispersion (-3.09 ps^2) and 151m-longer cavity. This is the first demonstration of coherently coupled PDW in a fiber laser using a bandpass filter and the formation of coherently coupled PDW is ascribed to the BPF's force filtering.

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1. Introduction

The formation of optical domain walls (DW) in fiber laser has captured the great interest attention recently. Both DW pulses and dark solitons are dark pulses that can be characterized as intensity dips embedded in a continuous wave background [1], in contrast to bright pulses. Dark solitons exist under normal group velocity dispersion, however, the formation of DW is irrelevant to dispersion [2]. There exist three kinds of DW in fiber laser currently: polarization domain walls (PDW) [3], color domain walls (CDW) [4–6], and polarization-color domain walls (PCDW) [7]. PDW refers to a localized structure that separates two mutually perpendicular polarization states. However, CDW and PCDW are coupled between different wavelengths. As for practical application, DW pulses have become an efficient way to generate the square-wave pulse and have been used in the field of data transmission [8–10].

PDW is theoretically proposed by V. E. Zakharov in nonlinear optics in 1987 [2], confirmed experimentally by Pitotis in 1998 [11]. Due to the existence of fiber birefringence, fiber can support two orthogonal degenerate modes. Cross-coupling between the two orthogonal polarization light components induces the formation of stable localized nonlinear structures which separate proximal domains of the two orthogonal polarization states of the optical field [12]. In 2009, H. Zhang et al. first experimentally observed coherently coupled PDW in a 14.7 m weakly birefringent Er-doped fiber ring laser [3]. Subsequently, they observed incoherently coupled PDW in a similar structure [13], whose polarization directions have obvious different wavelengths and spectral distributions. Besides, they theoretically analyzed coupled complex GinzburgLandau equation (CGLE) could well explain the generation of PDW pulses when the effect of spectral filtering on the pulse shaping could no longer be ignored. The formation of PDW is determined by intracavity birefringence-induced spectral filtering effect, which can be

described by $\Delta\lambda = \lambda^2/(\Delta n l)$ (Δn is net average cavity fiber birefringence) [13,14]. That is to say, only when the birefringence in the cavity is small enough can the coherent coupling (same central wavelengths) occur. And the coherent coupling is more likely to occur in short cavity weakly birefringent laser, the incoherent coupling often occur in longer cavity, especially with added highly nonlinear fiber. In 2012, Ning et al. experimentally observed incoherent coupling PDW in a 58 m figure-eight dispersion managed passively mode-locked fiber laser [15]. The relative separation of two different wavelength laser emissions could be altered when rotating PC or carefully bending the cavity fibers. In 2014, Tang et al. theoretically analyzed and experimentally observed the formation of PD between two linearly polarized beams co-propagating in a ~ 20 m weakly birefringent EDF laser [16]. By turning the wavelengths of the laser oscillations possibly close to each other (fine-tuning the PC), they experimentally observed incoherently and coherently coupled PDW pulses and theoretically studied their features based on the small parameter perturbation method. However, even in the case of a very weak birefringence, there always be slight differences in the spectra distribution of the two polarization components because there's more or less birefringence in fiber, although the center wavelength of the two polarization components remains the same. The coherent coupling has much stronger coupling strength and much weaker intensity noise than incoherent coupling [16], however, domain breaking could be easily observed under coherent coupling [17]. In 2020, Hu et al. observed the evolution of polarization domain splitting in a ~ 25 m Er-doped fiber ring laser under incoherent and coherent cross-polarization coupling with different group velocity dispersion [18]. Meanwhile, by using different kinds of saturable absorbers and mode locking cavity structures, some new results present the promising characteristics of DW pulses in Er-doped, Yb-doped, and Tm-doped fiber laser [8,14,19,20]. However, most of them are incoherently coupled [13,14,21,22], coherently coupled PDW pulses have seldom been reported [3,8,18]. Even under coherent coupling, PDW with same spectral distributions are rarely reported.

As for the research of PDW's multi-pulse state, it would give a new perspective to the understanding of PDW pulses' dynamics and intrinsic feature which may also find great potential applications in future optical communication systems. However, there are few reports on it so far. In 2019, Zhang et al. firstly observed up to 8th order mode-locking operation of PDW when the pump power is increased from 120 to 542.5 mW [23]. In 2020, Hu et al. studied the evolution of PDW splitting under different cavity dispersion, they found that the formed polarization domains can be more obvious and wide when the laser has relatively large average cavity dispersion [18]. In 2021, they went on to observe the coexistence and collision between soliton and polarization domain walls in fiber lasers under different cavity dispersion [24]. The formed domain patterns are stationary in the cavity and repeat with the cavity roundtrip time.

In this letter, we demonstrate the generation of stable optical PDW pulses in an Er-doped fiber ring laser. Rich temporal shapes of coherently coupled polarization domain walls based on a bandpass filter are experimentally observed firstly. Compared with past research, the cavity structure is more novel and shorter without using SA or other mode-locking structures and the pump laser is 976 nm. By fine adjusting the pump power and the polarization controllers, stable bright-dark pulse pairs, dark-bright pulse pairs, dark-bright-bright pulses, and dark-dark-bright pulses are observed. By increasing the length of the SMF in the cavity, we observed the multi-pulse pattern of the PDW pulse including randomly distributed PDW splitting, harmonic mode-locking, and periodic soliton collision. The Polarization-resolved study of the pulse dynamics indicates that coherent cross-coupling between the two orthogonal polarization components induces the formation of DW pulses.

2. Experimental setup

The proposed cavity configuration of the Er-doped fiber ring laser is shown in Fig. 1. A laser diode (LD) operating at 976 nm with maximum pumping power of 750 mW is employed to pump

5 m Er-doped fiber (EDF) through a 980/1550 nm wavelength-division multiplexer (WDM). The used Er-doped fiber has a core absorption coefficient of ≥ 3 dB/m at 980 nm, a core diameter of $3.2 \mu\text{m}$, and a numerical aperture of 0.23 with a group velocity dispersion (GVD) parameter of -12.2 ps/km/nm. A polarization controller (PC1) is used to control and optimize the polarization state of intra-cavity light. A polarization-independent isolator (PI-ISO) ensures the unidirectional operation of the laser.

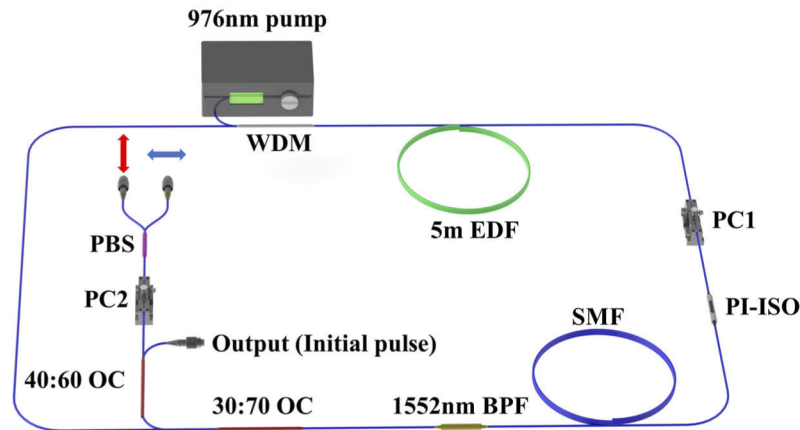


Fig. 1. Schematic of the Er-doped fiber (EDF) ring laser. WDM: wavelength division multiplexer, EDF: erbium-doped fiber, OC: optical coupler, PC: polarization controller, PI-ISO: polarization-independent isolator, BPF: bandpass filter, PBS: polarization beam splitter.

The key device to generate coherently coupled PDW pulses is a bandpass filter (BPF) with the center wavelength of 1552 nm and 0.5-dB bandwidth of 3 nm. The 30% port of a fiber output coupler (OC) is used to output the signal light and the 70% port is going back in the cavity. All the devices' pigtailed fiber is single-mode fiber. The total cavity length is ~ 8.6 m and the total dispersion in the cavity is -0.0002 ps². The single-mode fiber prior to BPF are used to adjust the net dispersion. In order to detect the output signal conveniently, an output signal is split by a 40:60 OC. The polarization-resolved measurement includes an OC, a PC2, and a polarization beam splitter (PBS), in which the 40% port of OC is used to observe the initial signal directly from the ring fiber, and 60% port of OC is used to separate two polarization directions through PBS to monitor the two orthogonal polarization components simultaneously. The PC2 located before the external PBS is used to balance the linear polarization change caused by the lead-fibers used outside of the cavity and find two Eigen linear polarization states of the laser emission.

An optical spectrum analyzer with a high resolution of 0.05 nm, a 2-GHz real-time oscilloscope with three 12.5 GHz InGaAs photodetector, and a radio frequency (RF) spectrum analyzer were used to observe the optical spectrum, temporal pulse shape, and RF spectrum, respectively.

3. Experimental results and discussions

3.1. Nearly zero negative dispersion

We keep the optical fiber its most natural state at first, where PC1 does not apply any pressure or torque to the fiber. And a typical output pulse train is observed at the pump power of 350 mW, as presented in Fig. 2(a), in which, we observe a pair of dark-bright pulse with unequal intensity in a period. The fundamental repetition frequency of the laser is 23.78 MHz, corresponding to the cavity length of 8.6 m. Figure 2(b) is zoom-in of Fig. 2(a). Initially, the total laser intensity output, the horizontal axis, and the vertical axis components are orthogonally polarized and

separated from the initial PDWs. The initial PDWs present a pair of bright and dark pulses embedded in continuous wave (CW) background, and it's obvious that PDW pulses coexist with CW consistently, which can be separated from the CW background by nonlinear Fourier transform as reported in [25]. At the location of polarization switching, it is characterized as the intensity alternation of the CW laser emissions between the two orthogonal polarization directions. The two orthogonal polarizations exhibit antiphase dynamics, when the CW oscillation along one polarization is on, the CW oscillation along the other is completely off, presenting complementary square pulses emission. A pair of bright-dark pulses appear at each switching position, while at the other switching position a weak intensity dip peak appears, as shown in Fig. 2(b). The differences in intensity are due to the intensity difference of edge before and after the polarized component pulse. The signal-to-noise ratio (SNR) is about 77 dB which demonstrates the high stability of the laser operation, as shown in Fig. 2(c). Sequentially, the PDWs' spectrum is illustrated in Fig. 2(d). In the spectral domain, all the wavelengths are centered at 1553.2 nm, the difference of their central wavelengths is less than 0.01 nm. Except for 3.4 dB differences in their spectral intensity induced by different optical paths, the two polarization components have a nearly uniform spectral distribution and the same central wavelengths, indicating that it is a coherent coupling [21,24].

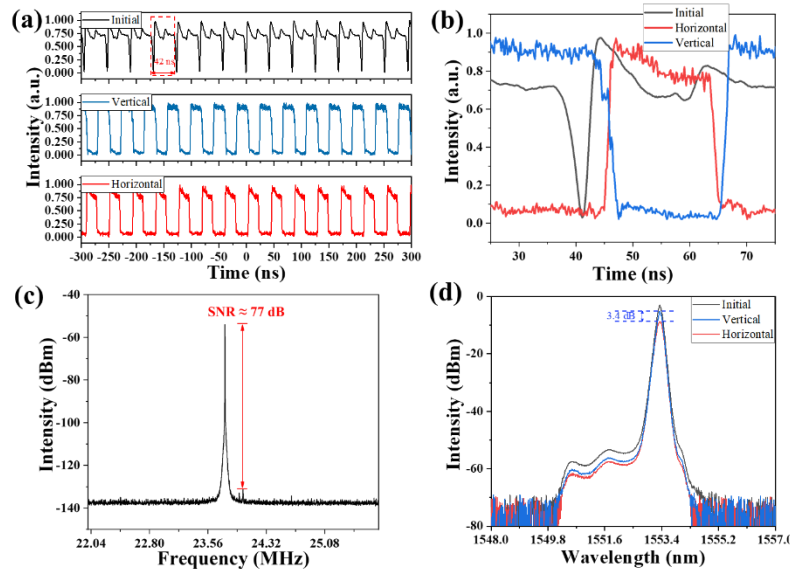


Fig. 2. A typical output pulses at pump power of 350 mW. (a) Long-range temporal trains before (initial) and after (horizontal axis and vertical axis) the PBS; (b) Short-range temporal trains; (c) RF spectrum of initial laser output at the fundamental repetition rate; (d) The corresponding optical spectrum

Rotating PC1 to a fixed position and keep it unchanged, we investigated experimentally the evolution of the state as the cross-polarization coupling strength is further increased. This is done by increasing the pump intensity. Thanks to the high strength of coherent coupling (small birefringence [13,20]), PDW pulse pairs are generated when the pump power is only 58 mW, which is much lower than other PDW pulse fiber lasers [21,23], as depicted in Fig. 3(a). Pulses characteristics and the output spectrum as a function of pump power are shown in Fig. 3(b) and Fig. 3(c), respectively. It can be found that the bright part is weakening with the intensity dips is narrower and deeper and the locations of polarization switchings keep stable. However, may be limited by the pump power, we didn't observe domain splitting as reported in [16,18]. The

result implies that the polarization domain formation does not need strong beams as far as the cross-polarization coupling is coherent.

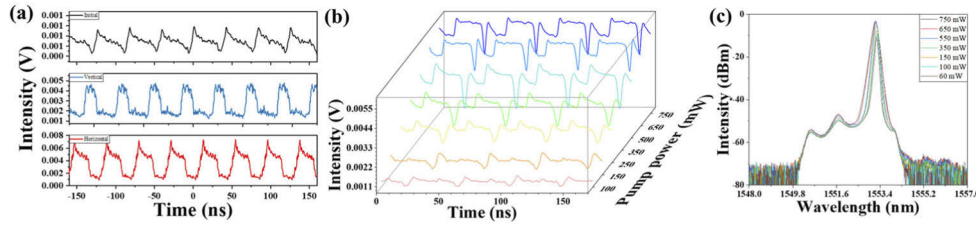


Fig. 3. Output pulse characteristics as a function of pump power. (a) Output pulses when the pump power is only 58 mW; (b) Output pulses as a function of pump power; (c) Output spectrum as a function of pump power.

Moreover, the shapes of the temporal pulses also varied with tuning intra-cavity PC. At fixed pump strength of 300 mW and by carefully tuning the orientation of the intra-cavity PC, we obtained a variety of temporal pulses, as shown in Figs. 4(a) to 4(f). As the strength of cavity birefringence increases, the duration of the laser emission in one polarization increases, in the meantime the duration in the orthogonal polarization decreases. This changes the relative position of the two domains in a period, thus displaying different initial pulse shapes. In Fig. 4(a), the Vertical polarization duration is long—27.8 ns, occupying $\sim 66\%$ of a period, and the two PDWs in one period are very close—14.2 ns, forming a dark-dark-bright pulse state. As the intracavity birefringence increased, the horizontal polarization duration increased from 14.2 ns to 29 ns, and the two PDWs gradually separated, forming dark-bright-bright pulse states, as shown in Figs. 4(b) to 4(d). In Figs. 4(e) and 4(f), the vertical polarization duration is 7 ns and 3 ns, so

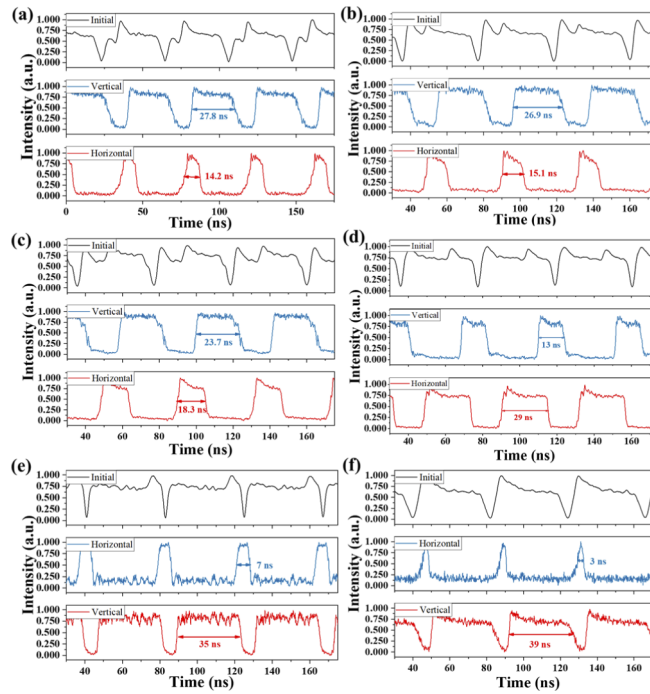


Fig. 4. Versatile patterns of temporal pulses with different polarization switching locations.

the two PDWs are superposed together again but with a slight change in their relative positions, dark-bright pulses and bright-dark pulses are formed respectively. However, as intracavity birefringence continues to increase or decrease based on Figs. 4(a) and 4(f), the laser emission becomes a continuous wave state. Nevertheless, this is the richest pulse shape to the best of our knowledge and the wide birefringence adjustment range indicates that our cavity has quite weak net birefringence.

Different from the previous literature, no matter how to tune the intra-cavity PC, there are few changes in the output pulse spectrum as shown in Fig. 5. The forced filtering introduced by the bandpass filter occurs stronger so that the birefringence filtering effect doesn't work. And since narrow filtering bandwidth makes spectral bandwidth of the formed pulses far narrower than the gain bandwidth, the effect of gain bandwidth spectral filtering on the pulse shaping could be ignored, the soliton should be governed by CGLE, PDW pulses can be formed [3,13]. Essential dissipative effects comprise gain saturation and bandpass filtering. Besides, cross-gain saturation between two laser modes is beneficial to the appearance of antiphase pulsations. Dissipative effects, accompanied by Kerr nonlinearities, make up CGLE, which can well explain the generation of PDW pulses [1,26].

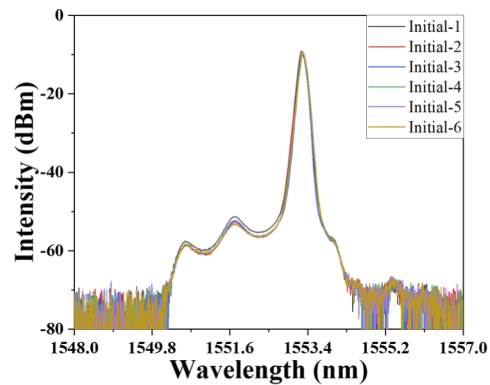


Fig. 5. Output spectrum corresponding to different temporal pulses

3.2. Larger negative dispersion

We added ~ 143 m SMF before BPF to increase the length of the cavity, introducing large birefringence and negative dispersion, and observed rich multi-pulse patterns of PDW. The repetition frequency drops to 1.35 MHz, and the total dispersion in the cavity is -3.09 ps². The spectrums and the temporal pulses at the pump power of 80 mW are shown in Fig. 6. The corresponding position of the peak spectrum is still at 1553.2 nm, and the spectral distribution changes slightly versus the short cavity's, which further indicates that the cavity is also a coherent coupled PDW pulse. As the length of the cavity increases, the random birefringence fluctuations increase, the birefringence filtering effect will become stronger, and the coherent coupling will likely become incoherent coupling, however, this cavity keeps coherent coupled in longer cavity, this is because the forced filtering caused by BPF occurs stronger than intracavity birefringence-induced filtering effect.

The pump threshold of PDW pulse pairs is only 25 mW due to the high accumulated nonlinearity of long cavity structure. When tuning PC1 with fixed pump power, we can observe the moving of polarization switching location. However, the duty cycle adjustment range in the long cavity is smaller than short cavity, occupying only one-sixth of a period, as shown in the inset arrows and panes of Fig. 6(b). This is because a long cavity increases the net birefringence of the cavity,

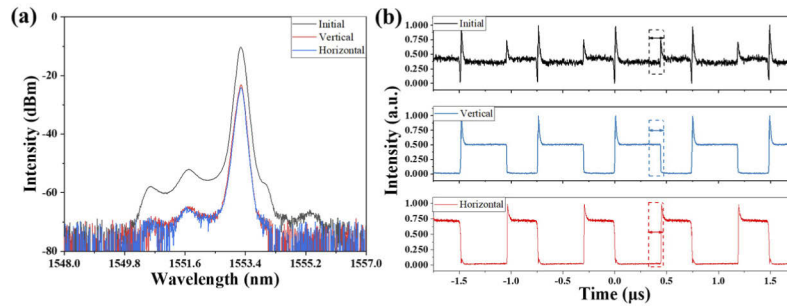


Fig. 6. Single-pulse operation under longer cavity. (a) The output spectrum; (b) The temporal pulses.

reducing the range of birefringence that can be adjusted so that the pulse evolution is not as rich as the short-cavity structure.

As the increase of the cross-polarization coupling strength (increasing the pump power to 142 mW), PDW pulses splitting occurs. It appears as an irregular multi-pulse state with rather irregular polarization component temporal pulse output. The temporal distribution of initial pulses in a period is different from time to time, as shown in [Visualization 1](#). The polarization switchings in a period occur more frequently and randomly, causing the randomly distributed multiple pulses of initial PDWs in a period. Figure 7 illustrates two snapshots of random distributed multi-pulse state. Although this PDW pulses state is random, it can be stationary in the cavity and repeat with the cavity roundtrip time.

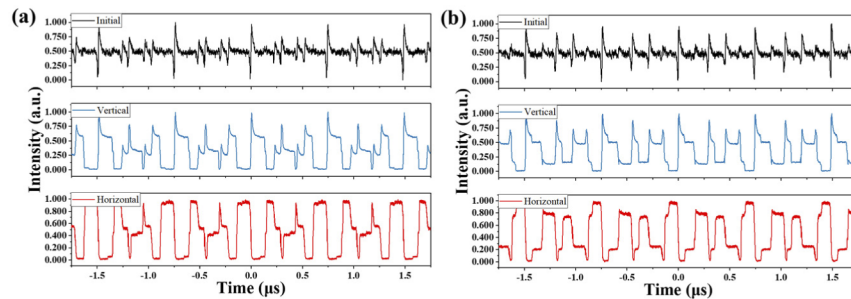


Fig. 7. Two snapshots of random distributed multi-pulse state.

By tuning PC1 carefully, random distributed multi-pulse state can turn into the stable harmonic mode-locking state, as shown in Fig. 8, and periodic soliton collision, as shown in [Visualization 2](#). Periodic soliton collision leads to the periodic moving of initial PDWs. Figure 9 illustrates the snapshots of the soliton collision operation. Note that, although the collisions lead to the perturbation on the pulses, the pulses would be eventually merged with the neighboring domain and disappear at last, owing to the self-organization effect [27,28]. Here, the spectral profile of different pulses at the same polarization may be inconsistent with each other, causing the different group velocities of two pulses because of the wavelength-dependent fiber dispersion. This is the reason why the two pulses move toward each other and collide periodically. It should be also noted that when the two pulses with different shapes were achieved, they could survive for a long time.

The long cavity structure greatly enhances the nonlinear effect, which helps to produce a multi-pulse state. Besides, the multi-pulse evolution is also associated with fine spectral filter

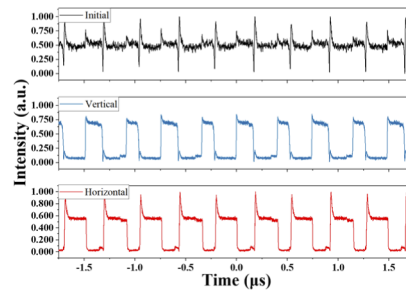


Fig. 8. Harmonic mode-locking pulses

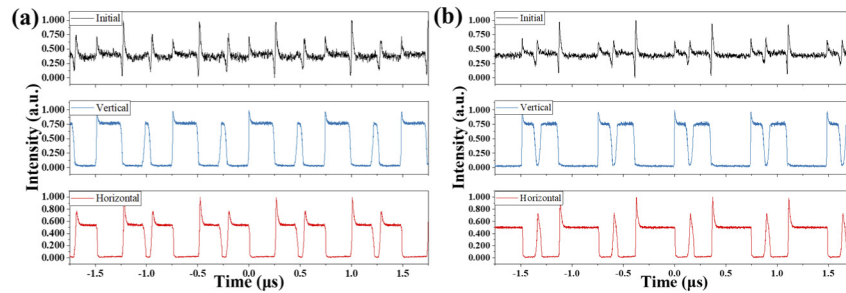


Fig. 9. Two snapshots of the soliton collision process

characteristics [29]. When removing the bandpass filter, regardless of the rotation of intra-cavity PC or changing of the pump power, we can only obtain a continuous wave at a center wavelength of 1558 nm, indicating that PDW pulses generation in this experiment relies largely on the spectral filter characteristics. Actually, under different net negative dispersion, we have observed the PDW pulses either, which shows that it is a general effect of the fiber lasers and has nothing to do with dispersion once again [17].

4. Conclusion

In conclusion, we have experimentally realized the stable output of completely coherently coupled PDW pulses in an Er-doped fiber ring laser based on a bandpass filter for the first time. By only adjusting the pump power and in-line PCs, a rich variety of pulse evolution with a wide birefringence adjustment range, including bright-dark pulse pairs, dark-bright-bright pulses, and dark-dark-bright pulses have been observed with the high SNR (77 dB) and low pump threshold (58 mW). By increasing the cavity length, we observed the multi-pulse patterns of PDW pulses. Polarization resolved measurement shows that two orthogonal polarization components have a wide square temporal profile and the same wavelength distribution. The formation and stable propagation of pulse pairs and dark pulses mainly result from the coherent cross-polarization coupling between the two orthogonal polarization components, which is attributed to the BPF's forced filter. Compared with past reviews, the cavity structure is shorter and more novel without using SA or other mode-locking structures, which shows again that PDW generation depends on fine cavity structure and is an essential feature of quasi-isotropic fiber lasers.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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